QAOA

The Quantum Approximate Optimization Algorithm

A step-by-step chapter: from graphs and cuts to a CUDA-Q implementation

What you will be able to do after this chapter

- compute cuts, optimal cuts and random-split statistics of small graphs quickly and reliably;
- translate Max-Cut between bits, spins, QUBO polynomials and the cost Hamiltonian, without sign errors;
- work with Pauli operators, rotation gates and CNOTs, including conjugation rules;
- write down the QAOA state and circuit, count its gates and parameters, and read the notebook's code;
- derive the $p = 1$ expectation value and use symmetries, light cones and conventions correctly;
- reason about optimizers, gradients, shot noise, backends, memory, noise and the limits of QAOA;
- solve every type of question in the accompanying 80-question QAOA assessment.

Companion material in the same repository: `presentations/QAOA_Lecture.pdf` (slides), `QAOA.ipynb` (CUDA-Q notebook) and `assessments/` (question paper and answer book).

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QAOA — The Quantum Approximate Optimization Algorithm

How to use this chapter

The chapter is built as a staircase. Sections 1–3 are the *mathematics* (graphs, cuts, probability, the Ising form). Sections 4–5 are the *quantum mechanics* (states, Pauli algebra, gates). Sections 6–9 are *QAOA itself* (state, circuit, $p = 1$ analysis, conventions). Sections 10–13 are the *advanced* material and the *implementation* (depth, optimizers, the notebook, backends, noise). Section 14 collects strategies and a map from every assessment question to the section that teaches it; Section 15 has practice problems with answers.

Each section has: the idea in words, the precise statement, a derivation, **worked examples** solved step by step, **common traps** (the wrong answers examiners like to offer) and a short checklist. Statements are marked as *general theory*, *the notebook's convention*, or *illustrative assumptions*; "previously recorded" results are the outputs stored in `QAOA.ipynb`.

Definition: Conventions used everywhere in this chapter

- Graph $G = (V, E)$, nodes $0, \dots, n-1$, weights w_{ij} (1 if unweighted), $W = \sum_{(i,j)\in E} w_{ij}$.
- A split is a bitstring $z = z_0 z_1 \dots z_{n-1}$, node 0 written first; $\bar{z}$ = all bits flipped; spin $s_i = (-1)^{z_i}$.
- $C(z) = \sum_{(i,j)} w_{ij}[z_i \neq z_j]$; **notebook:** $H_C = \sum_{(i,j)} \frac{w_{ij}}{2} Z_i Z_j$, $C = \frac{W}{2} - H_C$, $H_M = \sum_i X_i$.
- $|\boldsymbol{\gamma}, \boldsymbol{\beta}\rangle = e^{-i\beta_p H_M} e^{-i\gamma_p H_C} \cdots e^{-i\beta_1 H_M} e^{-i\gamma_1 H_C} |+\rangle^{\otimes n}$; $F_p = \langle C \rangle$; $r = F_p / C_{\max}$.
- $R_X(\theta) = e^{-i\theta X/2}$, $R_Z(\theta) = e^{-i\theta Z/2}$; two-qubit basis order $|00\rangle, |01\rangle, |10\rangle, |11\rangle$.

1 Graphs, cuts and Max-Cut

1.1 Graphs

A graph is a set of nodes joined by edges. The **degree** d_i of a node is its number of edges. Every edge has two ends, so

$$\sum_{i \in V} d_i = 2|E| \qquad \text{(handshake lemma)},$$

and therefore the number of odd-degree nodes is always even. A graph is **bipartite** if its nodes can be coloured with two colours so that every edge joins different colours; this is possible exactly when the graph has no **odd cycle**.

1.2 Cuts

Definition: Cut and Max-Cut

For a split z, the cut value is $C(z) = \sum_{(i,j)\in E} w_{ij}\,[z_i \neq z_j]$. Max-Cut asks for $C_{\max} = \max_z C(z)$ and an optimal z.

Because flipping all bits swaps the two group labels, $C(\bar{z}) = C(z)$: the 2^n bitstrings describe only 2^{n-1} different partitions, and optimal bitstrings always come in complementary pairs.

Key result: when can every edge be cut?

$C_{\max} = |E|$ (unit weights) $\iff$ G is bipartite $\iff$ G has no odd cycle. Every odd cycle leaves at least one of its edges uncut. Standard values: $C_{\max}(C_n) = n$ for even n and $n-1$ for odd n; $C_{\max}(K_n) = \lfloor n/2 \rfloor \lceil n/2 \rceil$; $C_{\max}(K_{a,b}) = ab$; Petersen graph: 12 of 15.

Worked example 1.1: the house graph

Square $0{-}1{-}2{-}3{-}0$ plus node 4 joined to 0 and 1.

Step 1. Bound: the triangle $0, 1, 4$ is odd, so $C \leq 6 - 1 = 5$.

Step 2. Construct: $z_0 z_1 z_2 z_3 = 0101$ cuts the 4 square edges; node 4 has one neighbour in each group, so exactly one roof edge is cut for either value of z_4. $C = 5$.

Step 3. Count: a 5-cut must cut all square edges (otherwise at most 4), so $z_0 z_1 z_2 z_3 \in \{0101, 1010\}$ and z_4 is free: 4 optimal bitstrings, 2 partitions.

Common trap

"A graph with an odd number of nodes cannot have all edges cut." False: a star with 7 nodes is bipartite. What matters is an odd *cycle*. Equally, do not confuse *bitstrings* (count 2^n) with *partitions* (count 2^{n-1}).

1.3 Complexity and classical algorithms

Max-Cut is NP-hard: no polynomial-time exact algorithm is known. *Minimum* cut, in contrast, is polynomial (max-flow/min-cut, Stoer–Wagner). The Goemans–Williamson algorithm (semidefinite relaxation plus random hyperplane rounding) guarantees $\mathbb{E}[C] \geq 0.878\, C_{\max}$ in polynomial time. Single-node-flip local search stops at a *local* optimum; there each node v has at least half its edges cut (flipping v changes the cut by $u_v - c_v \leq 0$), so $C \geq |E|/2$ — but the local optimum need not be global. The notebook uses brute force up to 22 nodes and local search with 200 restarts above.

Worked example 1.2: standard graph families

Step 1. C_9 (odd cycle): alternate colours round the cycle; the last edge closes an odd loop and stays uncut: $C_{\max} = 8$.

Step 2. K_5: with k nodes on one side, $k(5-k)$ edges are cut; $k = 2$ or 3 gives 6.

Step 3. $K_{3,3}$ is bipartite: all 9 edges cut. Petersen graph (15 edges, twelve 5-cycles): $C_{\max} = 12$ (brute force over 2^{10} splits).

Checklist

Odd cycle present? Bitstrings (2^n) or partitions (2^{n-1})? Bound first, then construct a split that reaches the bound.

2 Random splits and sampling statistics

Key result: random-split baseline

If every z_i is an independent fair bit, each edge is cut with probability $\frac{1}{2}$, so

$$\mathbb{E}[C] = \frac{W}{2}, \qquad \mathrm{Var}(C) = \sum_e \frac{w_e^2}{4} \ (= |E|/4 \text{ for unit weights}).$$

The variance formula uses that the edge indicators $z_i \oplus z_j$ are *pairwise* independent, even for edges sharing a node.

Consequences: $C_{\max} \geq W/2$ for every graph, and random guessing already has approximation ratio $(W/2)/C_{\max}$. For the notebook's graph: $W/2 = 15$, $\sigma = \sqrt{7.5} = 2.74$, $C_{\max} = 25$ is 3.7σ above the mean, and only $4/2^{15} = 1.22 \times 10^{-4}$ of random splits are optimal.

2.1 Random graphs

In $G(n, q)$ each of the $\binom{n}{2}$ pairs is an edge with probability q: $|E| \sim \mathrm{Binomial}(\binom{n}{2}, q)$. For $n = 15$, $q = 0.3$: mean 31.5, standard deviation $\sqrt{105 \cdot 0.21} = 4.70$. The seed-42 instance has 30 edges.

2.2 Shots

Method: estimates from shots

- A ± 1 observable such as $Z_i Z_j$ has variance $1 - \mu^2 \leq 1$: standard error $\leq 1/\sqrt{S}$. A 0/1 indicator has variance $\leq \frac{1}{4}$.
- A general observable with spread σ: standard error $\sigma/\sqrt{S}$.
- With success probability P per shot: number of successes in S shots is $\mathrm{Binomial}(S, P)$, $\sigma = \sqrt{SP(1-P)}$; $P(\text{none in } k) = (1-P)^k$; shots for confidence c: $S \geq \ln(1-c)/\ln(1-P)$.

Worked example 2.3: the recorded $P_{\mathbf{opt}} = 0.06$

Expected optimal shots in 4000: 240 (exactly what was recorded), $\sigma = \sqrt{4000 \cdot 0.06 \cdot 0.94} = 15.0$. $P(\text{none in } 10) = 0.94^{10} = 0.539$. For 99 % confidence: $S \geq \ln 0.01 / \ln 0.94 = 74.4$, i.e. 75 shots.

Worked example 2.4: variance of a random cut

For a uniformly random split each edge indicator $X_e = z_i \oplus z_j$ is a fair bit, and two indicators are uncorrelated even when the edges share a node (given z_j, the bits z_i and z_k are independent). Hence $\mathrm{Var}(C) = |E|/4$. For the notebook's graph: $30/4 = 7.5$, $\sigma = 2.74$; the optimum 25 lies $(25 - 15)/2.74 = 3.7\,\sigma$ above the mean.

Checklist

Mean $W/2$, variance $|E|/4$ (unit weights); standard error $\sigma/\sqrt{S}$; "none in k shots" $= (1-P)^k$.

3 From bits to spins: QUBO and the cost Hamiltonian

3.1 Two identities

$$[a \neq b] = a + b - 2ab \quad (a, b \in \{0,1\}), \qquad [z_i \neq z_j] = \frac{1 - s_i s_j}{2} \quad (s = (-1)^z).$$

The first gives the QUBO form, e.g. for the path 0–1–2: $C = z_0 + 2z_1 + z_2 - 2z_0z_1 - 2z_1z_2$ (the linear coefficient of z_i is its weighted degree). The second gives the Ising form:

$$C(z) = \frac{W}{2} - \sum_{(i,j)} \frac{w_{ij}}{2} s_i s_j.$$

3.2 The cost Hamiltonian

Z has eigenvalue s_i on $|z_i\rangle$, so $Z_iZ_j\,|z\rangle = s_is_j\,|z\rangle$ and

Key result: cost Hamiltonian (notebook convention)

$$H_C = \sum_{(i,j)\in E} \frac{w_{ij}}{2} Z_iZ_j, \qquad C = \frac{W}{2}\, I - H_C, \qquad E_0(H_C) = \frac{W}{2} - C_{\max}.$$

H_C is diagonal; its ground states are the optimal splits. The code (`cost_hamiltonian()`) builds exactly this, without the constant.

Worked example 3.5: QUBO to Ising

$f = 2x_0x_1 - x_0 - x_1$ with $x = (1-s)/2$: $2x_0x_1 = \frac{1}{2}(1 - s_0 - s_1 + s_0s_1)$, $-x_0 - x_1 = -1 + \frac{1}{2}(s_0 + s_1)$; the linear terms cancel: $f = \frac{1}{2}(s_0s_1 - 1) = -[x_0 \neq x_1]$.

Worked example 3.6: the triangle is frustrated

$H_C = \frac{1}{2}(Z_0Z_1 + Z_1Z_2 + Z_0Z_2)$: $+\frac{3}{2}$ on $000, 111$ and $-\frac{1}{2}$ on the other six strings. All three products cannot be -1 since their product is $s_0^2 s_1^2 s_2^2 = 1$. So $E_0 = -\frac{1}{2}$ (six-fold) and $C_{\max} = 2$.

Common trap

Sign errors: large cuts are *low* energy. $E_0 = W/2 - C_{\max}$, not $-C_{\max}$, and not $+(\dots)$.

Worked example 3.7: the cut polynomial of a path

Path 0–1–2: $C = (z_0 + z_1 - 2z_0z_1) + (z_1 + z_2 - 2z_1z_2) = z_0 + 2z_1 + z_2 - 2z_0z_1 - 2z_1z_2$. Check $z = 010$: $0 + 2 + 0 = 2$ (both edges cut). The linear coefficient of z_i is the degree of node i.

Checklist

$[z_i \neq z_j] = \frac{1}{2}(1 - s_is_j)$; $C = \frac{W}{2} - H_C$; large cuts are *low* energies; $E_0(H_C) = \frac{W}{2} - C_{\max}$.

4 Qubits, measurement and Pauli algebra

4.1 States and measurement

n qubits: $|\psi\rangle = \sum_z a_z |z\rangle$, $\sum |a_z|^2 = 1$; measuring returns z with probability $|a_z|^2$. For a diagonal observable $\langle H\rangle = \sum_z |a_z|^2 H(z)$; per edge $\langle C_{ij}\rangle = \frac{1}{2}(1 - \langle Z_i Z_j\rangle) = P(\text{edge cut})$. For a product state $\langle Z_0 Z_1\rangle = \langle Z_0\rangle\langle Z_1\rangle$ with $\langle Z\rangle = P(0) - P(1)$.

Global phases are unobservable; complex conjugation of all amplitudes leaves all probabilities unchanged. A state vector needs 16×2^n bytes in fp64 (8×2^n in fp32): 32 qubits fit on an 80 GB GPU in fp64, 33 in fp32.

4.2 Pauli algebra

$X^2 = Y^2 = Z^2 = I$; $XY = iZ$, $YZ = iX$, $ZX = iY$; reversing the order flips the sign. Pauli strings commute iff they anticommute at an even number of positions: $Z_0 Z_1$ commutes with $X_0 X_1$ but not with X_0, X_1 or $Y_0 Z_1$. Tensor products multiply factor-wise: $(Z \otimes Z)(X \otimes X) = (ZX) \otimes (ZX) = -Y \otimes Y$.

Key result: exponential of a Pauli

If $P^2 = I$: $e^{-i\phi P} = \cos\phi\, I - i \sin\phi\, P$. Conjugation rotates by twice the angle:

$$e^{i\beta X} Z e^{-i\beta X} = \cos 2\beta\, Z + \sin 2\beta\, Y.$$

Worked example 4.8: expected cut from amplitudes

One edge, $|\psi\rangle = \frac{1}{2}|00\rangle + \frac{\sqrt{3}}{2}|01\rangle$: $P(00) = \frac{1}{4}$, $P(01) = \frac{3}{4}$, $\langle Z_0 Z_1\rangle = \frac{1}{4} - \frac{3}{4} = -\frac{1}{2}$, so $\langle C\rangle = \frac{1}{2}(1 + \frac{1}{2}) = \frac{3}{4}$ — simply the probability that the edge is cut.

Worked example 4.9: products of Pauli strings

$(Z\otimes Z)(X\otimes X) = (ZX)\otimes(ZX) = (iY)\otimes(iY) = -Y\otimes Y$; in the other order $(XZ)\otimes(XZ) = (-iY)\otimes(-iY) = -Y\otimes Y$: $Z\otimes Z$ and $X\otimes X$ commute (two anticommuting positions).

Checklist

Probabilities are squared moduli; global phases are invisible; count anticommuting positions to decide commutation.

5 Gates: rotations and CNOT

$R_X(\theta) = e^{-i\theta X/2}$, so $R_X(2\beta) = e^{-i\beta X}$ and $R_X(2\beta)|0\rangle = \cos\beta\,|0\rangle - i\sin\beta\,|1\rangle$ (bit flip with probability $\sin^2\beta$). $R_Z(\theta) = \mathrm{diag}(e^{-i\theta/2}, e^{i\theta/2})$ changes only phases; after a Hadamard, $P(0) = \cos^2(\theta/2)$ for $HR_Z(\theta)|+\rangle$.

Key result: CNOT conjugation and the ZZ rotation

CNOT (control i, target j): $X_i \to X_i X_j$, $Z_j \to Z_i Z_j$, Z_i and X_j unchanged. Hence

$$\mathrm{CNOT}_{ij}\, R_Z^{(j)}(\theta)\, \mathrm{CNOT}_{ij} = e^{-i\theta Z_i Z_j/2} = \mathrm{diag}(e^{-i\theta/2}, e^{i\theta/2}, e^{i\theta/2}, e^{-i\theta/2}).$$

Worked example 5.10: entangling power of the cost gate

$e^{-i\gamma Z\otimes Z/2}\,|{+}{+}\rangle$ has amplitudes $\frac{1}{2}e^{\mp i\gamma/2}$; concurrence $2|c_{00}c_{11}-c_{01}c_{10}| = |\sin\gamma|$: maximal at $\gamma=\pi/2$, zero at $\gamma\in\pi\mathbb{Z}$.

Worked example 5.11: a bit flip by the mixer

$R_X(2\beta)\,|0\rangle = \cos\beta\,|0\rangle - i\sin\beta\,|1\rangle$: for $\beta=\pi/8$ the bit flips with probability $\sin^2(\pi/8) = \frac{2-\sqrt{2}}{4} = 0.146$.

Worked example 5.12: CNOT conjugation table

P	$\mathrm{CNOT}_{0\to1}\,P\,\mathrm{CNOT}_{0\to1}$
$X\otimes I$	$X\otimes X$
$I\otimes X$	$I\otimes X$
$Z\otimes I$	$Z\otimes I$
$I\otimes Z$	$Z\otimes Z$

X flows from control to target, Z from target to control. The last row is the reason why CNOT–R_Z–CNOT is a ZZ rotation.

Checklist

Half-angle convention: $R_X(2\beta)=e^{-i\beta X}$, $R_Z(\gamma w)\to e^{-i\gamma\frac{w}{2}ZZ}$ inside the CNOT pair.

6 The QAOA state

Definition: depth-p QAOA (notebook convention)

$$|\boldsymbol{\gamma},\boldsymbol{\beta}\rangle = \prod_{l=p}^{1} e^{-i\beta_l H_M} e^{-i\gamma_l H_C}\;|+\rangle^{\otimes n}\,, \qquad F_p = \langle\boldsymbol{\gamma},\boldsymbol{\beta}|C|\boldsymbol{\gamma},\boldsymbol{\beta}\rangle\,, \qquad r = F_p/C_{\max}.$$

Layer 1 acts first. The notebook stores `params` $=[\gamma_1,\dots,\gamma_p,\beta_1,\dots,\beta_p]$ (blocked); the kernel reads `gamma = params[layer]`, `beta = params[layers + layer]`.

6.1 The ingredients

- $|+\rangle^{\otimes n} = H^{\otimes n}\,|0\rangle^{\otimes n}$: uniform, $\langle C\rangle = W/2$. It is the eigenstate of H_M with the *largest* eigenvalue $+n$. If all $\gamma_l=0$, the mixers only add global phases and $F_p = W/2$.
- $e^{-i\gamma H_C}$ is diagonal: $|z\rangle\mapsto e^{-i\gamma H_C(z)}\,|z\rangle$. All Z_iZ_j commute, so it factorises exactly over the edges. It writes the cut value into the phase and changes no probability.
- $e^{-i\beta H_M} = \bigotimes_i R_X(2\beta)$: no entanglement; it makes neighbouring splits interfere. $[H_M,H_C]\neq 0$, so $e^{-i\beta H_M}e^{-i\gamma H_C}\neq e^{-i(\beta H_M+\gamma H_C)}$.

6.2 Why it should work: the adiabatic picture

Slow evolution keeps a system in the same eigenstate position. Trotterising a slow interpolation gives alternating layers with special angles, so $F_p^*\to C_{\max}$ as $p\to\infty$; QAOA optimises the angles

instead. Since $|+\rangle^{\otimes n}$ is the *top* state of $+H_M$, the matching target is the top state of the final Hamiltonian; to reach the ground state of H_C interpolate to $-H_C$, i.e. choose γ and β of **opposite sign** in the notebook's convention.

Worked example 6.13: the notebook's start point

The default "ramp" (`linspace(0.2, 0.6, p)` for γ, `linspace(0.6, 0.2, p)` for β) gives, for $p = 2$, $\boldsymbol{\gamma} = (0.2, 0.6)$, $\boldsymbol{\beta} = (0.6, 0.2)$ — both positive. An exact calculation (NumPy and CUDA-Q agree) gives $\langle C \rangle = 9.18 < 15$: worse than random. COBYLA recovers; the best angles found ($\boldsymbol{\gamma} \approx (0.367, 0.690)$, $\boldsymbol{\beta} \approx (1.117, -0.275)$, and $1.117 \equiv 1.117 - \pi/2 = -0.454$ by the $\pi/2$ period) have opposite signs.

Worked example 6.14: no cost phases, no progress

If every $\gamma_l = 0$, each mixer acts on its own eigenstate: $e^{-i\beta H_M} |+\rangle^{\otimes n} = e^{-i\beta n} |+\rangle^{\otimes n}$. The state stays uniform and $F_p = W/2$ (15 for the notebook's graph) for every $\boldsymbol{\beta}$.

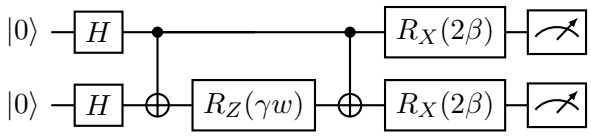
Worked example 6.15: the period in β

$e^{-i\frac{\pi}{2} H_M} = (-iX)^{\otimes n} = (-i)^n X^{\otimes n}$ flips every bit. It commutes with H_C and H_M, and $C(\bar{z}) = C(z)$, so shifting any β_l by $\pi/2$ leaves F_p unchanged: search β only in an interval of length $\pi/2$.

Checklist

$|+\rangle^{\otimes n}$ is the *top* eigenstate of $+H_M$; cost layer = phases only; mixer = interference; $\gamma\beta < 0$ heads for large cuts.

7 The circuit, gate counts and the code



One edge, one layer. In general every layer has, per edge, CNOT–$R_Z(\gamma w)$–CNOT on (control i, target j) and then $R_X(2\beta)$ on every qubit.

Key result: counts

gates $= n + p(3|E| + n)$; CNOTs $= 2p|E|$; R_Z: $p|E|$; R_X: pn; parameters $2p$. Default $(15, 30, 2)$: 225 gates, 120 CNOTs (recorded: `h 15, cx 120, rz 60, rx 30`, depth 64).

Common trap

The factor 2 in `rx(2.0 * beta)` and the absence of $\frac{1}{2}$ in `rz(gamma * w)` both come from the half-angle convention $e^{-i\theta P/2}$; they are not part of H_M or H_C.

7.1 The objective and the optimizer in the notebook

The objective calls `cudaq.observe(kernels["state"], H_C, ...).expectation()`, records $\langle C\rangle = W/2 - \langle H_C\rangle$ and returns $\langle H_C\rangle$ to `scipy.optimize.minimize` (COBYLA, `maxiter = 80`, `rhobeg = 0.2`: the initial trust-region radius). Minimising $\langle H_C\rangle$ is maximising $\langle C\rangle$. The final `cudaq.sample` uses the *best evaluated* angles (`argmax(history_cut)`), not necessarily `opt.x`; every sample's cut is computed classically.

Worked example 7.16: gate counts and parameter layout

Step 1. $n = 20$, $|E| = 57$, $p = 3$: gates $20 + 3(171 + 20) = 593$, of which $2 \cdot 3 \cdot 57 = 342$ CNOTs.

Step 2. $p = 3$, `params = [a, b, c, d, e, f]`: layer 2 is `layer = 1`: γ_2 = `params[1]` $= b$, β_2 = `params[3 + 1]` $= e$.

Step 3. Ramp for $p = 5$: $\gamma = (0.2, 0.3, 0.4, 0.5, 0.6)$, $\beta = (0.6, 0.5, 0.4, 0.3, 0.2)$ (`linspace` includes both end points).

Checklist

$n + p(3|E| + n)$ gates, $2p|E|$ CNOTs, $2p$ parameters; parameters are *blocked* $[\boldsymbol{\gamma}, \boldsymbol{\beta}]$; the best evaluated angles are sampled.

8 Exact analysis at $p = 1$

Key result: one edge

$\langle Z_0 Z_1\rangle = \sin 4\beta \sin\gamma$, $F_1 = \frac{1}{2}(1 - \sin 4\beta \sin\gamma)$. Maximum 1 at $(\frac{\pi}{2}, -\frac{\pi}{8})$; at $(\frac{\pi}{2}, \frac{\pi}{8})$ the edge is *never* cut.

Derivation. Mixer: $Z_0 Z_1 \to (cZ_0 + sY_0)(cZ_1 + sY_1)$ with $c = \cos 2\beta$, $s = \sin 2\beta$. Cost: $Z_0 Z_1$, $Y_0 Y_1$ unchanged; $Y_0 Z_1 \to \cos\gamma\, Y_0 Z_1 + \sin\gamma\, X_0$ and $Z_0 Y_1 \to \cos\gamma\, Z_0 Y_1 + \sin\gamma\, X_1$. In $|{+}{+}\rangle$ only X_0, X_1 survive (value 1): $\langle Z_0 Z_1\rangle = 2cs\sin\gamma$.

Key result: triangle-free graphs (Wang et al. 2018, translated to the notebook's sign)

$$\langle Z_u Z_v\rangle = \tfrac{1}{2}\sin 4\beta \sin\gamma \left(\cos^{d_u - 1}\gamma + \cos^{d_v - 1}\gamma\right).$$

Ring ($d = 2$): $\langle C_{uv}\rangle = \frac{1}{2} - \frac{1}{4}\sin 4\beta \sin 2\gamma \le \frac{3}{4}$: $r = \frac{3}{4}$. 3-regular graphs: $r \ge 0.6924$ (Farhi et al.).

Worked example 8.17: the path 0–1–2 at $(\frac{\pi}{2}, -\frac{\pi}{8})$

Edge $(0, 1)$: $d_0 = 1$, $d_1 = 2$: $\langle Z_0 Z_1\rangle = \frac{1}{2}(-1)(1)(1 + 0) = -\frac{1}{2}$ (note $\cos^0\gamma = 1$). Both edges alike: $\langle C\rangle = 2 \cdot \frac{1}{2}(1 + \frac{1}{2}) = 1.5$, $r = 0.75$. State vector: $P(010) = P(101) = 0.3125$, $P_{\text{opt}} = 0.625$.

Worked example 8.18: the triangle is solved at $p = 1$

H_C has two eigenvalues, so the cost layer sets a single relative phase between "bad" $(000, 111)$ and "good" strings. At $\gamma^* = \arctan(1/\sqrt{2})$, $\beta^* = -\gamma^*/2$ one finds $F_1 = 2 = C_{\max}$ exactly (numerically verified).

Worked example 8.19: the ring at $p = 1$

For a ring, $d_u = d_v = 2$: $\langle Z_u Z_v \rangle = \frac{1}{2} \sin 4\beta \sin \gamma \, (2 \cos \gamma) = \frac{1}{2} \sin 4\beta \sin 2\gamma$ and $\langle C_{uv} \rangle = \frac{1}{2} - \frac{1}{4} \sin 4\beta \sin 2\gamma \leq \frac{3}{4}$, reached at $(\gamma, \beta) = (\frac{\pi}{4}, -\frac{\pi}{8})$. Even ring C_8: $F_1 = 8 \cdot \frac{3}{4} = 6$, $r = 6/8 = 0.75$.

Worked example 8.20: one edge, traced amplitude by amplitude

$\gamma = \frac{\pi}{2}$, $\beta = \frac{\pi}{8}$ (notebook convention): after $H \otimes H$ all amplitudes are $\frac{1}{2}$; the cost layer gives $|00\rangle, |11\rangle$ the phase $e^{-i\pi/4}$ and $|01\rangle, |10\rangle$ the phase $e^{+i\pi/4}$ (relative phase 90°); after the mixer the probabilities are $(0.5, 0, 0, 0.5)$: the edge is *never* cut. With $\beta = -\frac{\pi}{8}$ it is always cut.

Checklist

Edge formula with degrees; $\cos^0 \gamma = 1$ for leaves; $F_1 \leq \frac{3}{4}$ per edge on rings; check signs of γ and β.

9 Symmetries and sign conventions

- $\beta_l \to \beta_l + \frac{\pi}{2}$ inserts $e^{-i\frac{\pi}{2} H_M} = (-i)^n X^{\otimes n}$ (flip all bits), which commutes with H_C and H_M and leaves $|+\rangle^{\otimes n}$ invariant: F_p has period $\frac{\pi}{2}$ in each β_l.
- $(\boldsymbol{\gamma}, \boldsymbol{\beta}) \to (-\boldsymbol{\gamma}, -\boldsymbol{\beta})$ gives the complex-conjugate state (real Hamiltonians and start state): same probabilities, same F_p.
- $H_C \to H_C + cI$ multiplies the state by a global phase.

Common trap

Textbooks often write $e^{-i\gamma C}$. Since $C = \frac{W}{2} - H_C$, $e^{-i\gamma C} = e^{-i\gamma W/2} e^{+i\gamma H_C}$: the notebook's circuit with $\gamma \to -\gamma$. Literature optima such as $(\frac{\pi}{2}, \frac{\pi}{8})$ become $(\frac{\pi}{2}, -\frac{\pi}{8})$. The constant is harmless; the sign is not.

Worked example 9.21: translating literature angles

A paper using $e^{-i\gamma C}$ quotes the single-edge optimum $(\gamma, \beta) = (\frac{\pi}{2}, \frac{\pi}{8})$. In the notebook's convention ($e^{-i\gamma H_C}$, $H_C = \frac{W}{2} - C$) the same circuit needs $(-\frac{\pi}{2}, \frac{\pi}{8})$ — or, using $(\gamma, \beta) \to (-\gamma, -\beta)$ symmetry, $(\frac{\pi}{2}, -\frac{\pi}{8})$. Using the literature angles unchanged gives $F_1 = 0$.

Checklist

Constant shifts → global phase (harmless); sign of H_C → sign of γ (not harmless); β period $\pi/2$; point symmetry $(\boldsymbol{\gamma}, \boldsymbol{\beta}) \to -(\boldsymbol{\gamma}, \boldsymbol{\beta})$.

10 Depth, light cones and limits

In $U^\dagger Z_u Z_v U$ every cost layer adds the neighbours of the current support; after p layers only nodes within distance p matter (a ring edge: $2p+2$ qubits). Consequences: fixed-p QAOA is local; for rings with $n \geq 2p+2$ the ratio is independent of n; numerically $r_p = \frac{3}{4}, \frac{5}{6}, \frac{7}{8}$ for $p = 1, 2, 3$, following $\frac{2p+1}{2p+2}$. With optimal angles $F^*_{p+1} \geq F^*_p$ (set the extra angles to 0) and $F^*_p \to C_{\max}$; the *found* value can still drop with a finite budget.

No quantum advantage for Max-Cut has been demonstrated. $p = 1$ guarantees 0.6924 on 3-regular graphs, below Goemans–Williamson's 0.878, and locality limits low-depth QAOA on some graph families (Bravyi et al. 2020).

Worked example 10.22: light cone on a ring

Edge (u, v), $p = 3$: the support grows $\{u, v\} \to 4 \to 6 \to 8$ nodes (one on each side per cost layer). So $\langle Z_u Z_v \rangle$ involves $2p+2 = 8$ qubits, and any ring with $n \geq 8$ gives the same per-edge value. Numerically the optimal ring ratios are $\frac{3}{4}, \frac{5}{6}, \frac{7}{8}$ for $p = 1, 2, 3$.

Checklist

Light cone $2p+2$ on rings; $F^*_{p+1} \geq F^*_p$ but found values may not improve; 0.6924 (QAOA $p = 1$, 3-regular) < 0.878 (Goemans–Williamson).

11 Optimizers, gradients and noise in the objective

- **Parameter shift.** For a single gate $e^{-i\theta P/2}$, $f(\theta) = a + b\cos\theta + c\sin\theta$ and $f' = \frac{1}{2}[f(\theta + \frac{\pi}{2}) - f(\theta - \frac{\pi}{2})]$ exactly. A shared angle (γ_l appears in $|E|$ gates) needs the chain rule: $2|E|$ runs per γ_l, $2n$ per β_l; $2p(|E|+n) = 180$ for the default. A two-term shift of γ_l as a whole is not exact.

- **Finite differences** (L-BFGS-B): $2p+1$ runs per gradient, approximate.

- **COBYLA**: derivative-free trust region; local, no global guarantee. The recorded run stopped at its budget (80); a re-run stopped after 75 evaluations at the trust-region lower bound, with identical results to 4 decimals.

- **Shot noise**: standard error $\sigma_C/\sqrt{S}$ (0.079 for $\sigma_C = 2.5$, $S = 1000$). On hardware the notebook always uses shots.

- **Better starts**: INTERP interpolates depth-p optima, e.g. $(0.4, 0.8) \to (0.4, 0.6, 0.8)$; parameter transfer reuses angles across graphs with similar local structure.

Worked example 11.23: gradient and shot budgets

Step 1. Exact parameter shift, gate by gate, for $n = 15$, $|E| = 30$, $p = 2$: $2p(|E|+n) = 180$ runs per gradient; forward differences: $2p+1 = 5$.

Step 2. Shot-based objective, spread $\sigma_C = 2.5$, $S = 1000$: standard error $2.5/\sqrt{1000} = 0.079$.

Step 3. INTERP from $\boldsymbol{\gamma}^{(2)} = (0.4, 0.8)$: $\boldsymbol{\gamma}^{(3)} = (0.4, \frac{1}{2}(0.4+0.8), 0.8) = (0.4, 0.6, 0.8)$.

Checklist

Count circuit runs per optimiser step; shot noise $\sigma/\sqrt{S}$; COBYLA's `rhobeg` is a trust-region radius, not a learning rate.

12 Results of the notebook and how to read them

	previously recorded	re-executed (WSL laptop)
COBYLA evaluations	80	75
best $\langle C\rangle$, r	20.5336, 0.8213	20.5336, 0.8213
best sampled cut	25	25
P(optimal split) (4000 shots)	0.0600	0.0600
execution cell	40.8 s	183.9 s

- r is an average; 6 % of shots (not 82 %) were optimal. The exact $P(C = 25)$ of the optimised state is 0.058 (lecture script).
- Random guessing already gives $r = 0.6$; the normalised improvement is $\eta = (20.53-15)/(25-15) = 0.553$.
- The four most frequent bitstrings are the four optimal ones; complementary pairs have nearly equal counts (68/67, 56/49).
- Timing: 40.01 s /80 $\approx$ 0.50 s per evaluation $\approx$ 2.2 ms per gate (calibration predicted 2.18 ms). Scaling to $n = 25$, $|E| = 90$: $\approx 2.8 \times 10^3\times$ longer on a CPU.
- Only the CPU state-vector path (`qpp-cpu`) has been executed; GPU, multi-GPU and hardware paths are implemented but unvalidated.

Worked example 12.24: reading the recorded run

Step 1. $r = 20.5336/25 = 0.821$, but random guessing already gives $15/25 = 0.6$: $\eta = (20.5336 - 15)/(25 - 15) = 0.553$.

Step 2. Partition $\{$`110010001001101`, `001101110110010`$\}$: $(68 + 67)/4000 = 0.0338$; the other optimal partition $(56 + 49)/4000 = 0.0263$; total 0.06.

Step 3. Time: 40.01 s/80 = 0.50 s per evaluation, 0.50/225 = 2.2 ms per gate (the calibration predicted 2.18 ms).

13 Backends, memory, noise and variants

Backends (`resolve_backend`): `cpu+statevector` $\to$ `qpp-cpu`; `cpu+density_matrix` $\to$ `density-matrix-cpu` (16×4^n B); `gpu+statevector` $\to$ `nvidia`; `gpu+mps` $\to$ `tensornet-mps`; `cpu+mps` $\to$ error (needs a GPU); `multi-gpu` needs `mpiexec`, one rank per GPU; 36 qubits in fp64 (1.1 TB) need 16 GPUs of 80 GB. MPS is efficient when few edges cross each cut of the qubit order ($\chi \le 2^{p\, m_k}$): rings yes, dense graphs no.

Noise (illustrative global depolarising model): $\langle C\rangle_{\text{noisy}} = F\langle C\rangle + (1 - F)\frac{W}{2}$, $F = (1 - \varepsilon)^{N_{\text{CNOT}}}$; default with $\varepsilon = 0.5\,\%$: $F = 0.548$, $\langle C\rangle \approx 18.0$. Limited connectivity adds SWAPs (3 CNOTs each).

Variants: warm-start QAOA (initial state from a relaxation), ADAPT-QAOA (mixers from an operator pool), Quantum Alternating Operator Ansatz (feasibility-preserving mixers such as XY), recursive QAOA (eliminate variables using $\langle Z_i Z_j\rangle$). None is in the notebook.

Worked example 13.25: memory and noise estimates

Step 1. 36 qubits in fp64: $16 \times 2^{36} = 1.10$ TB; with power-of-two slices of 80 GB GPUs: $2^4 = 16$ GPUs (68.7 GB each).

Step 2. Global depolarising model, 120 CNOTs, $\varepsilon = 0.005$: $F = 0.995^{120} = 0.548$; $\langle C \rangle = 0.548 \cdot 20.53 + 0.452 \cdot 15 = 18.0$.

Checklist

State memory $16 \cdot 2^n$ B (fp64); density matrix $16 \cdot 4^n$ B; MPS good for 1D-like graphs; noise pulls $\langle C \rangle$ towards $W/2$.

14 Problem-solving strategies and question map

14.1 Strategy by question format

- **Single-correct numerical:** compute fully, then sanity-check: cuts between 0 and $|E|$, $\langle C \rangle \geq W/2$ is *not* guaranteed, probabilities in $[0, 1]$. Typical distractors: "bitstrings instead of partitions", "variance instead of standard deviation", "textbook sign of γ", "one CNOT per edge".
- **Matching lists:** fix the easiest row first; one row often decides the option.
- **Assertion–Reason:** judge A and R independently; then ask whether R *causes* A (odd *cycles*, not odd node counts).
- **Statement I/II:** one counterexample kills a "for every" statement; read code statements against the code.
- **Paragraph:** reuse the quantity computed in the first question.

14.2 Fast checks that catch most mistakes

1. Is $C(\bar{z}) = C(z)$ respected (counts come in pairs)?
2. Does an odd cycle forbid the claimed cut?
3. Which convention: $e^{-i\gamma H_C}$ or $e^{-i\gamma C}$?
4. Average (r) or best sample (P_{opt})?
5. Recorded or re-executed numbers?

Method: five habits

1. **Convention first**: which sign ($e^{-i\gamma H_C}$ or $e^{-i\gamma C}$)? Which bit is written first? Blocked or interleaved parameters?
2. **Bound, then construct**: odd cycles bound $C_{\max}$; a 2-colouring of a large bipartite part achieves it.
3. **Count strings versus partitions**: factor 2 from $z \leftrightarrow \bar{z}$.
4. **Edge by edge**: $\langle C \rangle = \sum_e \frac{w_e}{2}(1 - \langle Z_u Z_v \rangle)$.
5. **Separate averages from best samples** and recorded from re-executed results.

Section	Assessment questions
1 Graphs, cuts	1, 2, 6, 7, 8, 14, 17, 18, 20, 78
2 Random splits, shots	3, 5, 10, 12, 13, 18, 21, 65, 79
3 QUBO, Ising, H_C	4, 9, 15, 19, 41, 53
4 States, Pauli algebra	22, 25, 26, 30, 31, 32, 35, 36, 38
5 Gates, CNOT	23, 24, 27, 28, 33, 39, 43, 44
6 QAOA state, ramp	11, 34, 42, 51, 52, 54, 55
7 Circuit, code	45, 46, 55, 58, 63, 64
8 $p = 1$ analysis	40, 47, 48, 56, 59, 60
9 Symmetries, conventions	29, 35, 37, 54
10 Depth, light cones, limits	49, 50, 61, 62, 77
11 Optimizers, gradients	16, 58, 63, 64, 65, 73, 74
12 Notebook results	13, 14, 21, 57, 75, 79, 80
13 Backends, noise, variants	32, 66, 67, 68, 69, 70, 71, 72, 76

15 Practice problems

Try each before looking at the answers on the next page. They use the same ideas as the assessment but different numbers.

P1. $C_{\max}$ of C_{11}, K_6, $K_{2,5}$.

P2. Expected cut and its variance for a random split of K_5 (unit weights).

P3. Write the cut of the star with centre 0 and leaves $1, 2, 3$ as a QUBO polynomial.

P4. Smallest eigenvalue of H_C for K_4 and its degeneracy (bitstrings).

P5. $\langle Z_0 Z_1 \rangle$ for the product state with $P(z_0 = 1) = 0.9$, $P(z_1 = 1) = 0.4$.

P6. Compute $e^{i\beta X} Y e^{-i\beta X}$.

P7. Gates and CNOTs of the notebook's circuit for $n = 10$, $|E| = 15$, $p = 4$.

P8. For `params = [a, b, c, d]` (so $p = 2$), give γ_2 and β_1.

P9. For one edge with weight $w = 3$ and $p = 1$: $F_1(\gamma, \beta)$ in the notebook's convention, and one optimal point.

P10. Ring C_{12}, $p = 1$, optimal angles: expected cut.

P11. The textbook optimum $(\gamma, \beta) = (0.6, 0.3)$ (convention $e^{-i\gamma C}$) in the notebook's convention.

P12. $P_{\text{opt}} = 0.01$: shots for 90 % confidence of at least one optimal sample.

P13. Exact parameter-shift runs for $n = 8$, $|E| = 12$, $p = 3$.

P14. Noise model: $N_{\text{CNOT}} = 200$, $\varepsilon = 0.002$, ideal $\langle C \rangle = 30$, $W/2 = 24$: noisy $\langle C \rangle$.

P15. Path $0{-}1{-}2{-}3$ at $p = 1$, $(\gamma, \beta) = (\frac{\pi}{2}, -\frac{\pi}{8})$: $\langle Z_u Z_v \rangle$ for the end edge and for the middle edge, and F_1.

P16. Number of exact parameter-shift runs for one gradient when $n = 6$, $|E| = 9$, $p = 2$, and the forward-difference count.

Answers to the practice problems

P1. 10 (odd cycle), $3 \cdot 3 = 9$, $2 \cdot 5 = 10$ (bipartite).

P2. $W/2 = 5$; Var $= 10/4 = 2.5$.

P3. $\sum_{k=1}^{3}(z_0 + z_k - 2z_0z_k) = 3z_0 + z_1 + z_2 + z_3 - 2z_0(z_1 + z_2 + z_3)$.

P4. $C_{\max}(K_4) = 4$, $W/2 = 3$: $E_0 = -1$; optimal strings have two nodes on each side: $\binom{4}{2} = 6$ bitstrings.

P5. $\langle Z_0 \rangle = 0.1 - 0.9 = -0.8$, $\langle Z_1 \rangle = 0.6 - 0.4 = 0.2$: -0.16.

P6. $e^{i\beta X}Y = Ye^{-i\beta X}$, so $Y(\cos 2\beta - i \sin 2\beta X) = \cos 2\beta\, Y - i \sin 2\beta\, YX = \cos 2\beta\, Y - \sin 2\beta\, Z$.

P7. $10 + 4(45 + 10) = 230$ gates; $2 \cdot 4 \cdot 15 = 120$ CNOTs.

P8. $\gamma_2 = b$, $\beta_1 = c$.

P9. Replace γ by $w\gamma$: $F_1 = \frac{w}{2}(1 - \sin 4\beta \sin 3\gamma)$; e.g. $(\gamma, \beta) = (\frac{\pi}{6}, -\frac{\pi}{8})$ gives $F_1 = 3$.

P10. $12 \times \frac{3}{4} = 9$ ($C_{\max} = 12$).

P11. $(-0.6, 0.3)$, or equivalently $(0.6, -0.3)$.

P12. $S \geq \ln 0.1 / \ln 0.99 = 229.1$: 230 shots.

P13. $2 \cdot 3 \cdot (12 + 8) = 120$.

P14. $F = 0.998^{200} = 0.670$; $0.670 \cdot 30 + 0.330 \cdot 24 = 28.0$.

P15. End edge ($d = 1, 2$): $\frac{1}{2}(-1)(1)(1 + 0) = -\frac{1}{2}$; middle edge ($d = 2, 2$): $\frac{1}{2}(-1)(1)(0 + 0) = 0$. $F_1 = 2 \cdot \frac{3}{4} + \frac{1}{2} = 2$ ($C_{\max} = 3$, $r = \frac{2}{3}$).

P16. $2 \cdot 2 \cdot (9 + 6) = 60$ parameter-shift runs; $2p + 1 = 5$ forward-difference runs.

Summary of the chapter in ten lines

1. $C(z) = \sum w[z_i \neq z_j]$; $C(\bar{z}) = C(z)$; 2^n strings, 2^{n-1} partitions.
2. Odd cycles block a perfect cut; bipartite $\iff$ $C_{\max} = |E|$; Max-Cut NP-hard, min cut easy.
3. Random split: mean $W/2$, variance $|E|/4$.
4. $[z_i \neq z_j] = \frac{1}{2}(1 - s_is_j) \Rightarrow C = \frac{W}{2} - H_C$, $H_C = \sum \frac{w}{2} Z_iZ_j$.
5. $e^{-i\phi P} = \cos\phi - i \sin\phi P$; conjugation rotates by 2ϕ; CNOT–$R_Z(\theta)$–CNOT $= e^{-i\theta ZZ/2}$.
6. $|\boldsymbol{\gamma}, \boldsymbol{\beta}\rangle = \prod e^{-i\beta_l H_M} e^{-i\gamma_l H_C} |+\rangle^{\otimes n}$; $n + p(3|E| + n)$ gates, $2p$ parameters.
7. One edge: $F_1 = \frac{1}{2}(1 - \sin 4\beta \sin\gamma)$; rings $\frac{3}{4}, \frac{5}{6}, \frac{7}{8}$; triangle exact at $p = 1$.
8. Period $\frac{\pi}{2}$ in β; $(\boldsymbol{\gamma}, \boldsymbol{\beta}) \to -(\boldsymbol{\gamma}, \boldsymbol{\beta})$ symmetric; textbook $\gamma = -$notebook γ.
9. Light cones make fixed-p QAOA local; $F_p^* \to C_{\max}$ only as $p \to \infty$.
10. Read results honestly: r vs. best sample vs. P_{opt}; recorded vs. re-executed; no speed-up claimed.